Marwadi University Marwadi Chandarana Group NAAC A+	**Marwadi University** **Faculty of Engineering &Technology**	
Subject: DSIP	**Aim:Simulate smoothing and sharpening operations on images using frequency domain filters.**	
Experiment:-10	**Date:30-10-2026**	**Enrollment No:92400133063**

Code:-

```
import cv2
import numpy as np
import matplotlib.pyplot as plt
# 1. Read image using image path
image_path = '/content/images.jpeg'
img = cv2.imread(image_path, 0)
if img is None:
    raise FileNotFoundError(f"Could not load the image at {image_path}. Please
verify the file path.")
# 2. Fourier Transform
F = np.fft.fftshift(np.fft.fft2(img))
M, N = img.shape
# 3. Create frequency distance matrix
u, v = np.meshgrid(np.arange(N)-N//2, np.arange(M)-M//2)
D = np.sqrt(u**2 + v**2)
# 4. Low-Pass and High-Pass Filters
LPF = D < 30
HPF = D >= 30
# 5. Apply filters
smooth = np.abs(np.fft.ifft2(np.fft.ifftshift(F * LPF)))
high_pass = np.abs(np.fft.ifft2(np.fft.ifftshift(F * HPF)))
sharp = np.clip(img.astype(float) + high_pass, 0, 255)
# 6. Display images
plt.figure(figsize=(12, 4))

plt.subplot(1, 3, 1)
plt.imshow(img, cmap='gray')
plt.title('Original Image')
plt.axis('off')
```

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```
plt.subplot(1, 3, 2)
plt.imshow(smooth, cmap='gray')
plt.title('LPF - Smoothening')
plt.axis('off')

plt.subplot(1, 3, 3)
plt.imshow(sharp, cmap='gray')
plt.title('HPF - Sharpening')
plt.axis('off')

plt.show()
```

output:-



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Conclusion:-

The code demonstrates frequency-domain image filtering using the 2D Fourier Transform. By computing the FFT of the image and applying a circular mask in the frequency domain, a low-pass filter (LPF) is used to smooth the image by preserving low-frequency components, while a high-pass filter (HPF) isolates high-frequency edges and details. The high-pass output is then added back to the original image to produce a sharpened result, enhancing edge clarity. This approach effectively illustrates how frequency separation enables targeted image enhancement — smoothing for noise reduction and sharpening for detail preservation — using simple radial distance thresholds in the Fourier spectrum.